

What agreement criterion does a reconstruction benchmark need?

A cross-environment reproduction of a low-dose CT evaluation

[**TODO:** author list, affiliations, ORCIDs, corresponding author]

September 14, 2026

Abstract

Purpose: Benchmarks in low-dose CT reconstruction increasingly ship a numerical agreement criterion so that a third party can verify a reported result. We asked what such a criterion must look like to survive a reproduction in a different computing environment, and whether a criterion that fails should be read as a failed reproduction.

Methods: An evaluation of five reconstruction methods over 764 held-out slices at three dose levels was recomputed by an independent operator, in a separate environment differing in operating system, CUDA build, and deep-learning framework version, from checkpoints and data verified by hash. Agreement was assessed against the benchmark’s pre-registered criterion, an absolute difference of 10^{-6} . Because that criterion failed, we tested the leading explanation — non-deterministic GPU kernels — by re-running the affected methods with deterministic kernels enabled, and separately audited the criterion itself across a ladder of absolute and relative tolerances.

Results: The pre-registered criterion failed for three of five methods. The determinism hypothesis was falsified: re-running the same side with deterministic kernels reproduced the original bit-for-bit, leaving the residual attributable to the environment rather than to kernel non-determinism. The criterion audit showed that no absolute tolerance resolves the comparison — it fails at 10^{-6} and equally at 10^{-3} — because the reported quantities span from a dimensionless index near 0.9 to an observer statistic near 10^6 . A relative criterion at 10^{-4} , declared per metric in that metric’s own units, admits every method, with a worst observed relative difference of 6.36×10^{-5} .

Conclusions: A single absolute tolerance cannot express agreement across metrics of differing scale, and loosening it does not repair it; only changing its kind does. We recommend that benchmarks declare agreement per metric in that metric’s own units, and that a criterion failure be investigated before it is either accepted as a discrepancy or relaxed.

Keywords: low-dose CT, reproducibility, benchmark, numerical agreement, detectability

1 Introduction

A benchmark that publishes a leaderboard implicitly promises that its numbers can be recomputed. Increasingly that promise is made explicit: a released evaluation ships a tolerance, and a recomputation that lands inside it is called a reproduction. The tolerance is rarely justified, and it is usually a single number.

We report a reproduction in which that construction failed, and in which the failure was informative rather than terminal. The pre-registered criterion of the benchmark under study was an absolute difference of 10^{-6} applied uniformly. Three of five methods fell outside it. The obvious reading — that the recomputation disagreed with the release — turned out to be wrong twice over: the difference was not caused by the mechanism everyone expects, and the criterion could not have been satisfied by any absolute threshold worth declaring.

The contribution of this note is not the verdict. It is the demonstration that a criterion’s *kind* matters more than its magnitude, and a worked procedure for establishing that by experiment rather than by preference.

2 Methods

2.1 The evaluation under reproduction

The benchmark evaluates five reconstruction methods — RED-CNN [1], CTformer [2], LEARN [3], CoreDiff [4], and a Gaussian blur control included deliberately as a smoothing baseline — over 764 held-out slices at three simulated dose levels. Each run reports fidelity (PSNR, SSIM) and task-based detectability (a channelized Hotelling observer AUC, a contrast-to-noise ratio, and a non-prewhitening observer with an eye filter). Summary statistics and the number of contributing slices give **[TODO: state the count: 5 methods $\times$ 3 dose levels $\times$ 5 reported quantities = 75 comparisons per method]** per method, 375 in total.

The five quantities differ in scale by roughly six orders of magnitude, which is the crux of what follows (Table 1).

Quantity	Units	Typical magnitude	Declared criterion
PSNR	dB	$\sim 40\text{--}54$	relative, 10^{-4}
SSIM	index in $[0,1]$	~ 0.9	relative, 10^{-4}
CNR	ratio	$\sim 2\text{--}7$	relative, 10^{-4}
CHO AUC	AUC in $[0,1]$	~ 1.0	relative, 10^{-4}
NPWE	observer statistic	$1.6 \times 10^5\text{--}9.7 \times 10^5$	relative, 10^{-4}
n	count	764	absolute, exact

Table 1: **The reported quantities and their natural scales.** A single absolute tolerance is simultaneously far too loose for a bounded index and far too tight for an observer statistic of order $10^5\text{--}10^6$. The declared criterion in the right-hand column is the one adopted after the audit in Section 2.5.

2.2 The two environments

The reproduction was performed by an operator independent of the original run, using a separate checkout, a separate virtual environment, and a separate output tree; no artifact of the original run was overwritten. The recomputation environment was Python 3.12.10 with PyTorch 2.3.0+cu121 and NumPy 1.26.4, on Windows with two NVIDIA RTX 4090 GPUs (driver 591.86). The reference environment differed in operating system, CUDA build, and framework version. **[TODO: Record the reference environment exactly — OS, driver, CUDA, torch, numpy. The recalculation report characterises it only as “the author’s environment”, which is not enough for a paper whose subject is cross-environment agreement. This is the single most important gap to close before submission.]**

This is a reproduction across environments, not a re-seed within one. That distinction is the point: a re-seed tests the pipeline’s internal determinism, whereas a different library stack tests whether the published number is a property of the method or of the machine.

2.3 Provenance control

Inputs were pinned by hash rather than by description: the source commit, with per-file SHA-256 verification of the six files carrying evaluation logic; the task specification; the five model checkpoints; and every file of both data trees (364 and 25 entries). The recomputation environment is recorded as a complete dependency freeze.

2.4 The determinism ablation

The standard explanation for small cross-run differences on GPU is non-deterministic kernel selection. This hypothesis is testable and it makes a sharp prediction: with deterministic kernels enabled, a re-run of the same side must be bit-identical, and any residual against the reference must therefore come from somewhere else.

We re-ran the three methods outside the pre-registered criterion under the following settings, at a cost of approximately 48 GPU-hours:

```

cudnn.deterministic = True
cudnn.benchmark = False
torch.use_deterministic_algorithms(True)
CUBLAS_WORKSPACE_CONFIG = :4096:8

```

2.5 The criterion audit

Independently of the verdict, we evaluated the comparison against a ladder of criteria: absolute and relative, each at 10^{-6} through 10^{-3} . The audit changes no measurement; it asks only which criteria the *same* recorded differences would satisfy.

3 Results

3.1 Agreement

Table 2 reports the comparison. Under the pre-registered absolute criterion, 115 of 375 comparisons differed. The blur control and LEARN reproduced bit-for-bit; RED-CNN, CTformer and CoreDiff did not.

The largest single disagreement is instructive. For RED-CNN the worst absolute difference is 34.3, which sounds disqualifying until one notes that it occurs in a quantity of magnitude $\sim 9 \times 10^5$ — a relative difference of 6.36×10^{-5} .

3.2 The determinism hypothesis is false here

The prediction of Section 2.4 failed. Re-running the recomputation side with deterministic kernels reproduced the non-deterministic run exactly: zero differing comparisons, bit-identical. The evaluation path was not sensitive to kernel non-determinism in this environment, so route (a) as scoped could not close the residual, and the difference against the reference is a systematic cross-environment offset.

Method	Comparisons	$n_{\neq}$ (abs. 10^{-6})	Worst relative difference (metric)	Outside declared
Blur (control)	75	0	bit-identical	0
LEARN	75	0	bit-identical	0
CTformer	75	25	4.41×10^{-7} , CNR	0
RED-CNN	75	45	6.36×10^{-5} , CNR	0
CoreDiff	75	45	2.50×10^{-5} , CNR	0

Table 2: **Agreement by method.** $n_{\neq}$ counts comparisons differing at all under the pre-registered absolute 10^{-6} criterion. The worst relative difference and its metric are given for each method; the final column counts comparisons outside the declared per-metric relative criterion of 10^{-4} , which is zero everywhere. That two methods reproduce exactly establishes that the pipeline, data and code snapshot are capable of bit-identical reproduction, so the differences in the other three are a property of those computations rather than of the harness.

This is the result that makes the rest of the note interpretable. Before it, environment and kernel non-determinism were confounded, and any change to the criterion would have been an adjudication made after seeing the outcome.

3.3 No absolute tolerance resolves the comparison

Table 3 is the central result. The pre-registered criterion fails at 10^{-6} ; it also fails at 10^{-5} , 10^{-4} and 10^{-3} , with the same three methods outside at every level. Loosening an absolute tolerance by three orders of magnitude changes nothing, because the binding comparison is on a quantity of order 10^5 – 10^6 , where an absolute budget of 10^{-3} is still a relative demand of order 10^{-9} .

A relative criterion behaves differently. It excludes two methods at 10^{-6} and 10^{-5} and admits all five at 10^{-4} .

Criterion kind	Tolerance	Verdict	Methods outside
Absolute	10^{-6}	FAIL	CTformer, RED-CNN, CoreDiff
Absolute	10^{-5}	FAIL	CTformer, RED-CNN, CoreDiff
Absolute	10^{-4}	FAIL	CTformer, RED-CNN, CoreDiff
Absolute	10^{-3}	FAIL	CTformer, RED-CNN, CoreDiff
Relative	10^{-6}	FAIL	RED-CNN, CoreDiff
Relative	10^{-5}	FAIL	RED-CNN, CoreDiff
Relative	10^{-4}	PASS	—
Relative	10^{-3}	PASS	—

Table 3: **The criterion ladder.** Applied to the same recorded differences, with no measurement changed. An absolute criterion cannot be repaired by loosening it; only changing its kind resolves the comparison.

We note what the audit also establishes in the other direction: the pre-registered criterion was not unattainable in principle. A 764-term float64 reduction accumulates a relative error of order 10^{-13} , so within a single environment the absolute 10^{-6} budget is comfortable. The criterion was not too strict; it was the wrong kind.

3.4 A provenance failure that numerical agreement would not have caught

The first reproduction attempt disagreed with the reference by 9–13 dB in PSNR for CTformer. The cause was not numerical: the operator guide named one checkpoint while the released run had used another, a retrained variant. Re-running with the checkpoint the release actually used restored agreement.

We report this because it is the failure mode a tolerance cannot detect. A criterion answers whether two numbers agree; it is silent on whether they are numbers about the same thing. Hash-level pinning of checkpoints, not their names, is what caught it.

4 Discussion

4.1 Declare agreement per metric, in that metric’s own units

The recommendation follows from Table 3 rather than from taste. A benchmark reporting quantities of differing scale should declare, for each, whether agreement is absolute or relative and at what tolerance, with counts required to match exactly. A single number applied across all of them is a statement about the largest-magnitude quantity and is vacuous for the rest.

4.2 A criterion failure is a finding, not a formality

The sequence matters, and we would put it forward as a procedure. The criterion failed; the leading explanation was stated as a falsifiable prediction; the prediction was tested at real cost and falsified; only then was the criterion amended, and the superseded criterion and its verdict were retained verbatim beside the new one. Had the criterion been relaxed at the first failure, the amendment would have been indistinguishable from adjusting a threshold until the result passed.

4.3 Limitations

Cross-environment agreement of a metric computation is not agreement of the underlying science. What reproduces here is a computation, on fixed inputs, under declared hashes. Nothing in this note speaks to whether the methods compared are clinically preferable, whether the detectability surrogate used measures diagnostic content, or whether the dose levels are attainable in practice.

The comparison involves two environments. A third would establish whether the offset is systematic across environments or particular to this pair, and this remains the clearest extension. **[TODO: Decide whether to add a third environment; the constraint is moving checkpoints and data under their use agreement, not compute.]**

The dose levels are simulated rather than separately acquired, and the patient counts of the underlying cohort are small; both limit what the reproduced numbers mean, though neither affects whether they reproduce.

5 Conclusions

A reproduction that fails its pre-registered criterion has not necessarily failed. In the case reported here the criterion was badly scaled, the mechanism usually blamed was absent, and the residual was a cross-environment offset four orders of magnitude inside a defensible relative tolerance. Benchmarks that ship an agreement criterion should declare it per metric, in that metric’s own units, and should treat its failure as the beginning of an investigation.

Data and code availability

[**TODO:** Point at the released artifact, the comparison JSON, the re-derivation script, and this note’s table generator, once the repository’s public release decision is made.]

Author contributions

[**TODO:** CRediT statement.]

Competing interests

[**TODO:** Per-author declaration.]

References

- [1] Hu Chen, Yi Zhang, Mannudeep K. Kalra, Feng Lin, Yang Chen, Peixi Liao, Jiliu Zhou, and Ge Wang. Low-dose CT with a residual encoder-decoder convolutional neural network. *IEEE Transactions on Medical Imaging*, 36(12):2524–2535, 2017. doi: 10.1109/TMI.2017.2715284.
- [2] Dayang Wang, Fenglei Fan, Zhan Wu, Rui Liu, Fei Wang, and Hengyong Yu. CTformer: Convolution-free Token2Token dilated vision transformer for low-dose CT denoising. *Physics in Medicine & Biology*, 2023.
- [3] Hu Chen, Yi Zhang, Yunjin Chen, Junfeng Zhang, Weihua Zhang, Huaiqiang Sun, Yang Lv, Peixi Liao, Jiliu Zhou, and Ge Wang. LEARN: Learned experts’ assessment-based reconstruction network for sparse-data CT. *IEEE Transactions on Medical Imaging*, 2018. arXiv:1707.09636.
- [4] Qi Gao, Zilong Li, Junping Zhang, Yi Zhang, and Hongming Shan. CoreDiff: Contextual error-modulated generalized diffusion model for low-dose CT denoising and generalization. *arXiv preprint arXiv:2304.01814*, 2023.